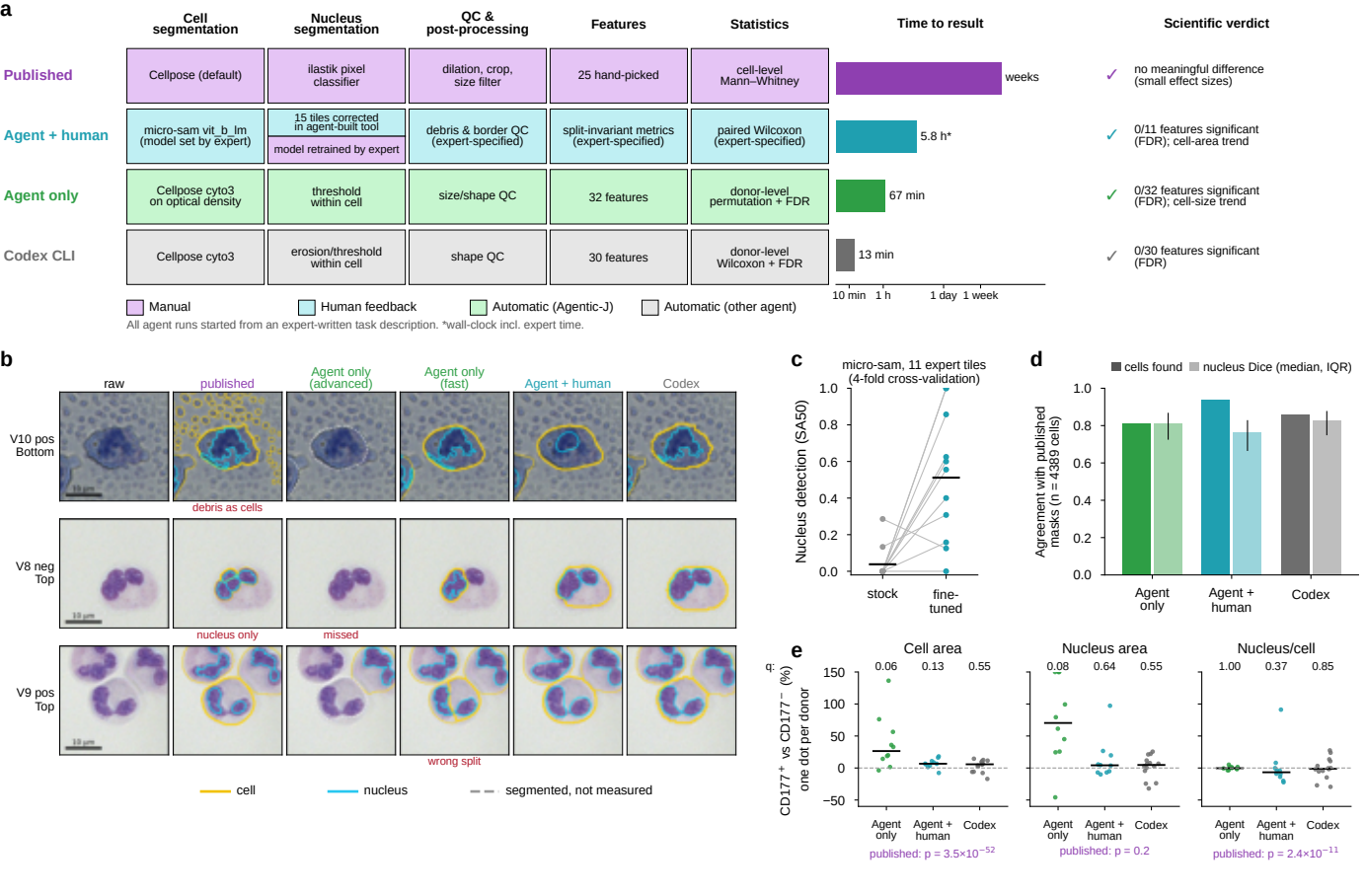

a
Cell segmentation
Nucleus segmentation
QC & post-processing
Features
Statistics
Time to result
Scientific verdict
Published
Cellpose (default)
ilastik pixel classifier
dilation, crop, size filter
25 hand-picked
cell-level Mann–Whitney
weeks
no meaningful difference (small effect sizes)
Agent + human
micro-sam vit_b_lm (model set by expert)
15 tiles corrected in agent-built tool
model retrained by expert
debris & border QC (expert-specified)
split-invariant metrics (expert-specified)
paired Wilcoxon (expert-specified)
5.8 h*
0/11 features significant (FDR); cell-area trend
Agent only
Cellpose cyto3 on optical density
threshold within cell
size/shape QC
32 features
donor-level permutation + FDR
67 min
0/32 features significant (FDR); cell-size trend
Codex CLI
Cellpose cyto3
erosion/threshold within cell
shape QC
30 features
donor-level Wilcoxon + FDR
13 min
0/30 features significant (FDR)
10 min
1 h
1 day
1 week
Manual
Human feedback
Automatic (Agentic-J)
Automatic (other agent)
All agent runs started from an expert-written task description. *wall-clock incl. expert time.
b
raw
published
Agent only (advanced)
Agent only (fast)
Agent + human
Codex
V10 pos Bottom
V8 neg Top
V9 pos Top
debris as cells
nucleus only
missed
wrong split
cell
nucleus
segmented, not measured
c
micro-sam, 11 expert tiles (4-fold cross-validation)
Nucleus detection (SA50)
stock
fine-tuned
d
cells found
nucleus Dice (median, IQR)
Agreement with published masks (n = 4389 cells)
Agent only
Agent + human
Codex
e
Cell area
Nucleus area
Nucleus/cell
q:
0.06
0.13
0.55
0.08
0.64
0.55
1.00
0.37
0.85
CD177+ vs CD177− (%) one dot per donor
published: p = 3.5×10−52
published: p = 0.2
published: p = 2.4×10−11